ESCUELA COLOMBIANA DE INGENIERÍA JULIO GARAVITO

UNIVERSIDAD

Design and Development of a Low-Cost GPS and Proximity Sensor-Based Mobility Assistance System for People with Visual Disabilities

Entity: Escuela Colombiana de Ingeniería Julio Garavito

Subject: ISIB_M - Ingeniería de Software para IoT y Big Data

Product Name: VisorSense

Product Type: Portable mobility assistance device for people with visual disabilities.

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1. Objective

Develop a highly affordable, scalable, and easy-to-implement technological mobility assistance solution using GPS positioning and proximity sensors to guide and alert people with visual disabilities during their daily journeys.

2. Target Group

- People who are totally blind or have low vision and seek greater safety and independence during their daily journeys.
- Foundations, universities, communities, and organizations interested in low-cost, easy-to-implement technological assistance solutions.

3. Metrics

- Detection of Accuracy: Percentage of obstacles correctly identified within the configured range.
- Response Time: Time elapsed between obstacle detection and the emission of the acoustic alert.
- Positioning Accuracy: Accuracy of the GPS system during navigation tests.
- Battery Life and Availability: Operating hours per charge and percentage of operational time without failures.

4. Overview

- Advanced mobility assistance solutions often involve high costs and complex technological requirements.

- The proposal consists of a low-cost portable device that integrates GPS-assisted navigation, obstacle detection through proximity sensors, and acoustic alerts.
- Processing is performed primarily locally on an ESP32 microcontroller or equivalent, reducing dependence on external infrastructure.

5. Product Description

- The device continuously obtains the user's position through GPS and locally processes measurements from proximity sensors to identify nearby obstacles.
- The sensors can be strategically positioned to cover frontal and lateral areas, allowing the detection of objects that may pose an impact risk during movement.
- The microcontroller evaluates the distance and location of obstacles and determines the level of risk without the need to transmit images or video to external services.
- A buzzer generates differentiated acoustic patterns to communicate obstacle proximity and basic navigation events to the user.

6. Vision

Facilitate the safe and independent mobility of people with visual disabilities through accessible, reliable, and low-cost technology that enables them to navigate and avoid obstacles in urban and rural environments.

7. Competitors

- Specialized commercial mobility assistance devices with high costs or complex technological requirements.
- Mobile navigation applications and solutions that primarily depend on a smartphone, connectivity, or external infrastructure.

8. Need Addressed

- Reduce the risk of collisions with obstacles and improve orientation during both familiar and unfamiliar journeys.
- Complement the use of the white cane, increase user independence, and provide an affordable alternative to specialized commercial devices.

9. Product

- A portable device based on an ESP32 or ESP32-S3 microcontroller that integrates GPS, proximity sensors, and an acoustic alert system.
- The GPS module obtains the user's position and enables basic navigation instructions based on a stored route or destination.
- Ultrasonic or infrared sensors detect frontal and lateral obstacles and nearby objects that may pose an impact of risk.
- Local processing determines the level of risk and generates different sound patterns according to the distance and location of the obstacle and navigation events.

- The solution is designed to be affordable, portable, repairable, and easily replicable by communities, universities, and foundations.

10. Business Objectives

- Design and assemble functional prototypes based on GPS and proximity sensors.
- Achieve an obstacle detection rate above 90% and response times below 500 ms during the tests defined for the prototype.
- Validate the solution with real users while maintaining a manufacturing cost significantly lower than that of existing commercial solutions.

11. Buyers

- Foundations supporting people with visual disabilities.
- Universities, inclusive education centers, and social organizations.
- Communities interested in manufacturing, adapting, and maintaining low-cost assistance solutions.

12. Key Activities

- Firmware development and integration of GPS and proximity sensors.
- Sensor calibration, navigation algorithm development, and energy optimization.
- Field testing and validation of safety, usability, and performance with users.
- Key Value Propositions
- Extremely affordable and easy-to-implement mobility assistance technology.
- Real-time guidance and alerts with local processing, low power consumption, and minimal dependence on external infrastructure.

13. Customer Relationships

- Co-creation, direct testing, and continuous validation with people with visual disabilities.
- Community support, accessible documentation, and assistance with manufacturing, maintenance, and adaptation.

14. Customer Segments

- People with visual disabilities require additional support for navigation and obstacle detection.
- Foundations, universities, and communities seeking to implement accessible technological assistance solutions.

15. Key Resources

- Hardware: ESP32, GPS module, ultrasonic or infrared sensors, buzzer, battery, and portable enclosure.
- Software: Embedded firmware, navigation algorithms, and obstacle detection algorithms.
- Technical knowledge in embedded systems, electronics, low-power IoT, and technological accessibility.

16. Channels

- Social media, demonstrations at universities and foundations, and outreach activities.
- Open Hardware repository, technical documentation, workshops, and a community of users and developers.

17. Cost Structure

- Cost of physical components, assembly, enclosure, battery, and prototype testing.
- Maintenance, component replacement, and development costs; the architecture minimizes recurring costs associated with cloud services.

18. Empathy Map

18.1. Thinks and Feels

- Fears of tripping, colliding with obstacles, or getting lost during journeys.
- Is concerned that a technological tool might fail when it is needed most.
- Wants greater safety and confidence when traveling through familiar and unfamiliar places.
- Wants to maintain independence without constantly relying on other people.

18.2. Sees

- Obstacles and environmental changes that may pose risks during journeys.
- People using phones, canes, or other tools for navigation.
- Urban and rural environments with different signage and traffic conditions.
- Specialized assistance devices that may be expensive or inaccessible.

18.3. Hears

- Instructions and recommendations from family members, companions, and other people.
- Environmental noise that can make it difficult to perceive external signals.
- Acoustic alerts from devices and navigation applications.
- Experiences and recommendations from other people with visual disabilities.

18.4. Says and Does

- Uses a white cane and other strategies to move around independently.
- Memorizes frequently traveled routes and requests guidance when the environment requires it.
- Tests technological navigation tools when he believes they can provide additional safety.
- Looks for solutions that are simple, reliable, and easy to incorporate into his routine.

19. User Persona: Enrique Ortiz

19.1. General Profile

Enrique Ortiz is a 55-year-old man who lives in Bogotá, Colombia. He is a retired accountant, single, and has one son. He became blind five years ago and currently uses a white cane as his primary mobility tool.

He is characterized as independent, cautious, and introverted. Although he sometimes travels accompanied by his son, he tries to avoid constantly depending on his family because he is concerned about becoming a burden to them.

19.2. Demographics

- Age: 55
- Gender: Male
- Place of residence: Bogotá, Colombia
- Occupation: Retired accountant
- Marital status: Single
- Children: 1
- Visual disability: Acquired blindness five years ago

19.3. Personality and Characteristics

- Independent and committed to maintaining his autonomy.
- Cautious when traveling, especially in unfamiliar places.
- Introverted and unlikely to ask for help.
- Values practical, accessible, and easy-to-use solutions.
- Tries not to constantly depend on family members or third parties.

19.4. Says

- “I prefer not to depend on someone accompanying me all the time.”

19.5. Thinks

- “If I encounter an unexpected obstacle, I need to receive an alert in time.”
- Enrique considers it important to have a technological tool that complements the use of the white cane and allows him to travel with greater safety, confidence, and independence.

19.6. Does

- Uses a white cane as his primary mobility tool.
- Travel independently whenever conditions allow.
- Sometimes travel is accompanied by his son.
- Ask other people for help only when the situation requires it.
- Tries to avoid situations in which he may feel dependent on other people.

19.7. Feels

- Relief and confidence when he successfully completes a journey independently and safely.
- Conversely, he feels concern and insecurity before traveling through unfamiliar places or when obstacles and environmental changes exist that may be difficult to identify in time.

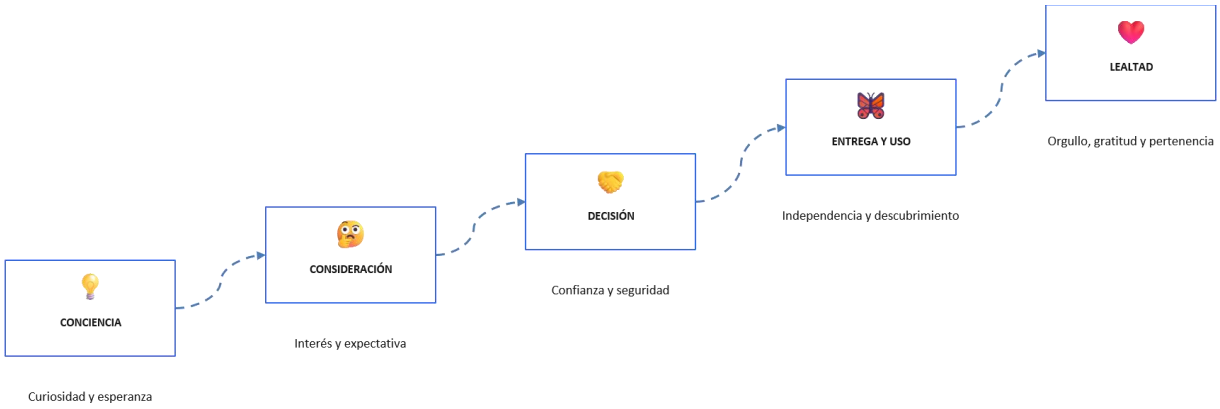
19.8. Goals

- Travel with greater safety and confidence.
- Maintain independence in everyday activities.
- Reduce the need to depend on family members or outside assistance.
- Have access to affordable technology that complements the white cane.
- Use a simple solution that adapts to his needs without replacing the mobility tools he already knows.

19.9. Pain Points

- Risk of accidents due to obstacles that are not detected in time.
- Difficulty navigating unfamiliar places or environments with changes.
- Dependence on third parties to resolve mobility-related situations.
- Specialized devices that are expensive, complex, or difficult to implement.

20. Customer Journey Map



21. System Architecture

21.1. Operating Flow

21.1.1. Initialization

- The device is powered on.
- The GPS obtains the current location.
- The sensors begin scanning the environment.

21.1.2. Continuous Monitoring

- The sensors constantly measure distances.
- The microcontroller evaluates potential risks.
- The direction and proximity of obstacles are determined.

21.1.3. Alert Generation

- Distant obstacle: slow beep.
- Nearby obstacle: rapid intermittent beep.
- Immediate risk: continuous beep.

21.1.4. Navigation

- The GPS calculates the location.
- The location is compared with the stored route or destination.
- Directional instructions are generated through sound patterns.

21.2. Hardware Components

- ESP32.
- NEO-6M GPS module.
- HC-SR04 ultrasonic sensor.
- Infrared sensor.
- Active buzzer.
- Rechargeable battery.
- Power switch.
- Enclosure.

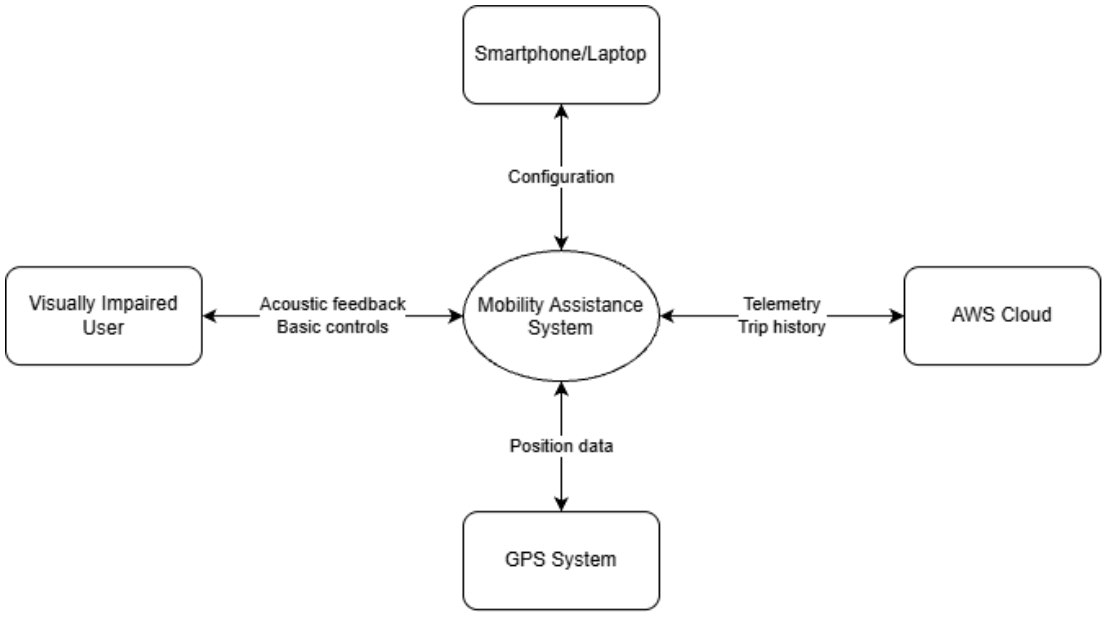
21.3. Functional Requirements

ID	Functional Requirement
FR-01	Detect frontal obstacles within a configurable distance.
FR-02	Detect lateral obstacles to reduce collision risks.
FR-03	Emit acoustic alerts according to obstacle proximity.
FR-04	Obtain geographic position using GPS.
FR-05	Provide basic navigation instructions.
FR-06	Operate without permanent Internet connectivity.
FR-07	Maintain low energy consumption for extended use.

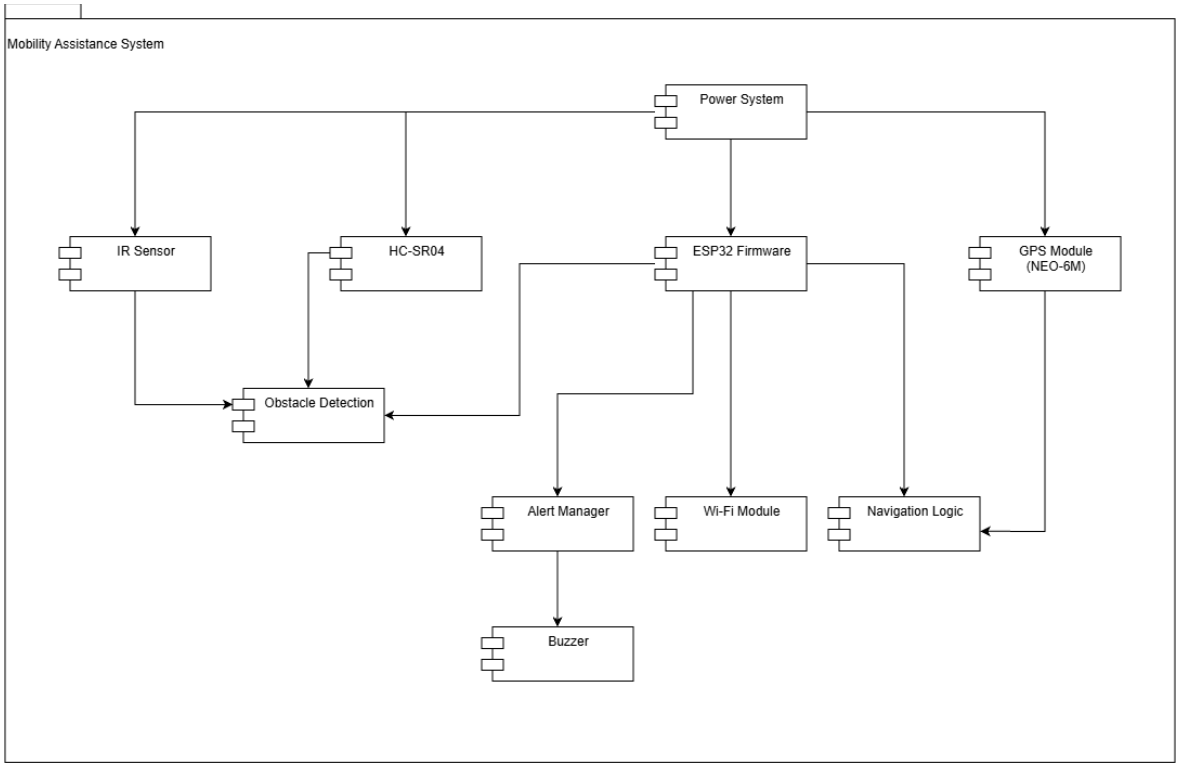
21.4. Risks and Limitations

- Limited GPS accuracy in densely populated urban areas.
- Ultrasonic sensors may be affected by certain surfaces.
- It does not completely replace tools such as the white cane.
- Proper calibration is required for different environments.
- Accuracy partially depends on environmental conditions.

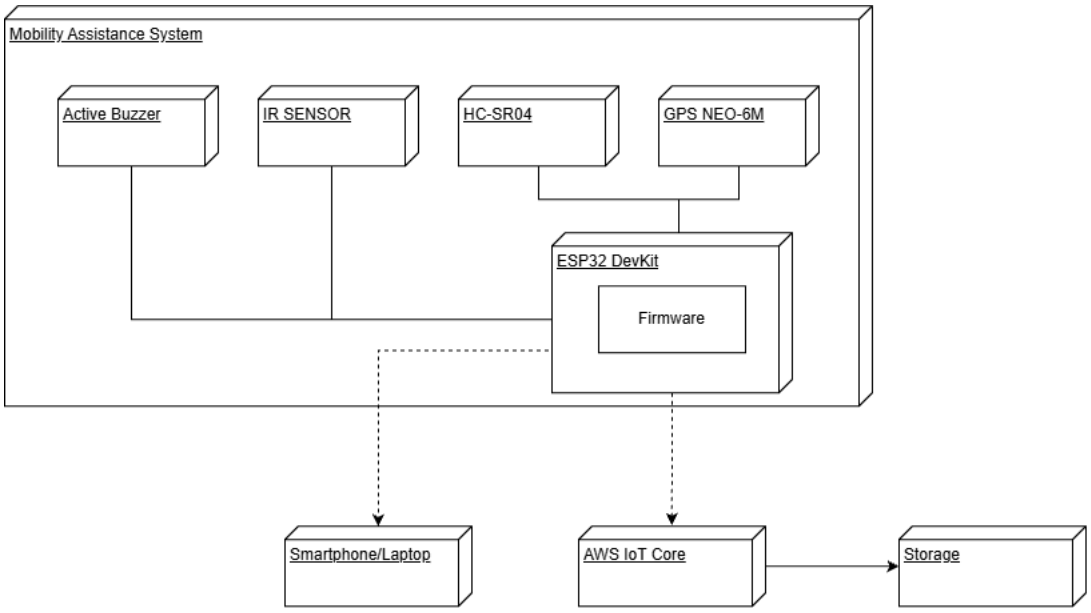
22. Context Diagram



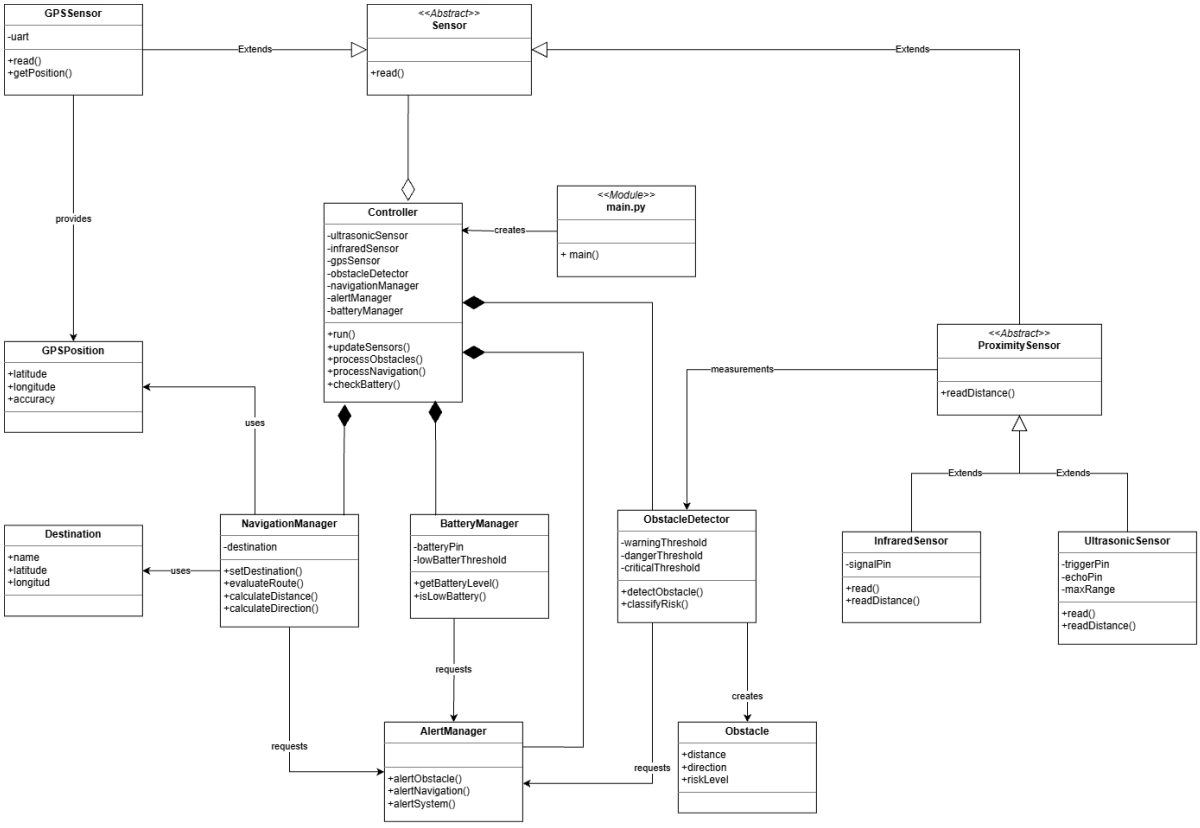
23. Component Diagram



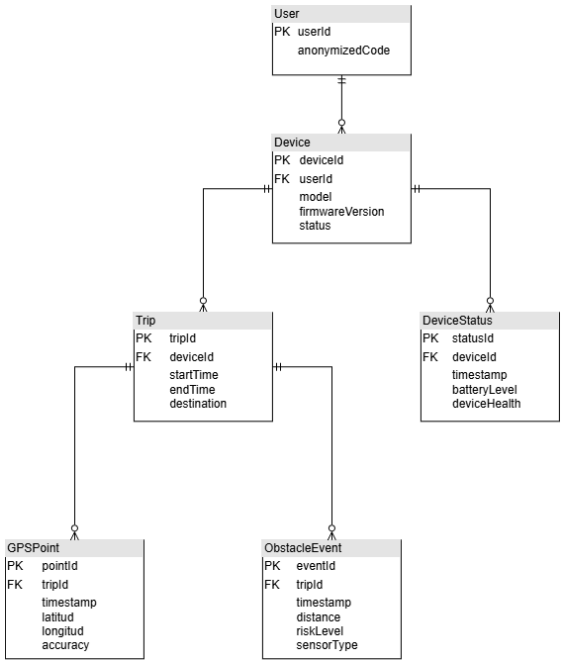
24. Deployment Diagram



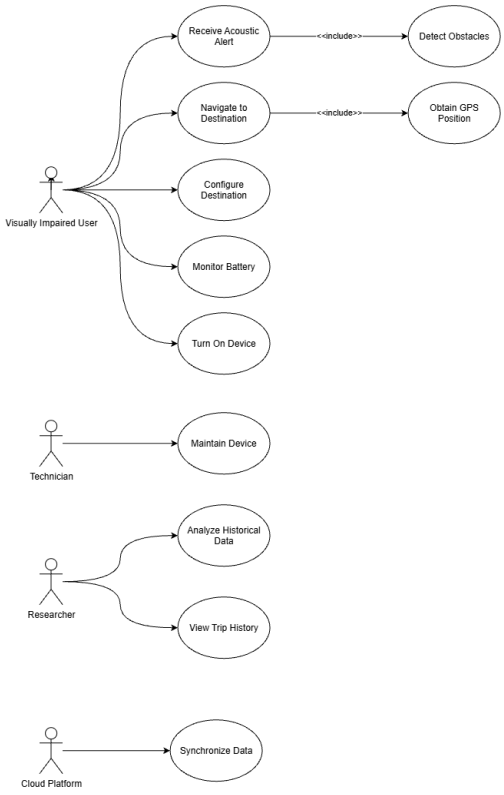
25. Class Diagram



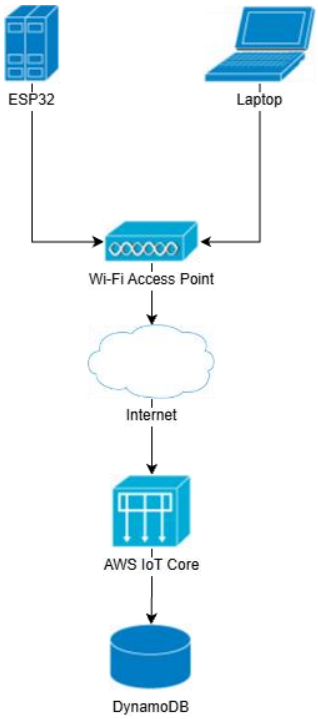
26. ER Diagram



27. Use Case Diagram



28. Cisco Network Topology



29. Electrical Block Diagram

29.1. Voltage Levels

29.1.1. 3.3V

- ESP32
- Most GPS NEO-6M modules (logic)
- LED
- Button

29.1.2. 5V

- HC-SR04
- Many IR modules
- Active buzzer (depending on model)

30. Conceptual Electrical Schematic

30.1. Hardware Assumptions

- ESP32 DevKit V1
- NEO-6M UART GPS
- HC-SR04 powered at 5V

- Active 5V buzzer
- IR obstacle sensor

30.2. Connection Table

Component	Pin	ESP32 GPIO
GPS TX	UART RX	GPIO16
GPS RX	UART TX	GPIO17
HC-SR04 TRIG	Digital Out	GPIO25
HC-SR04 ECHO	Digital In	GPIO26
IR OUT	Digital In	GPIO27
Buzzer Driver	Output	GPIO33
LED	Output	GPIO2
Button	Input	GPIO32

31. PCB / Connection Design

32. ESP32 Pinout

33. MVP Definition

33.1. Must Have

- ESP32
- GPS NEO-6M
- HC-SR04
- Buzzer
- Local alert logic
- Risk classification
- GPS positioning
- Battery power
- Functional testing

33.2. Should Have

- HC-SR04 + IR combination
- Destination management
- Wi-Fi synchronization
- AWS IoT Core
- Historical storage
- Dashboard